The Ambient AI Lab

AI for Scientific Practice

Will Handley
<wh260@cam.ac.uk>

Royal Society University Research Fellow
Institute of Astronomy, University of Cambridge
Kavli Institute for Cosmology, Cambridge
Gonville & Caius College
willhandley.co.uk/talks

7th September 2026

Scientists versus agents

Nature Astronomy Comments, 2026: Institute of Astronomy, Cambridge

Large language models are not the problem

Hiranya V. Peiris

Nature Astronomy, 3 April 2026

[doi:10.1038/s41550-026-02837-2] [2604.22071]

Agentic research is oxymoronic

Natalie B. Hogg

Nature Astronomy, 31 August 2026

[doi:10.1038/s41550-026-02954-y] [2608.31161]

ANTHROPIC

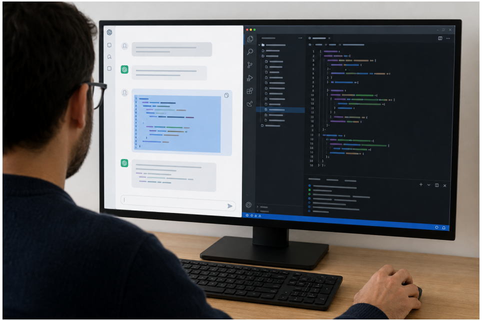
DeepMind

OpenAI

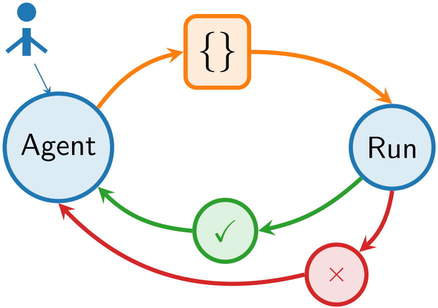
An agent is ChatGPT that can edit files and run commands

This closes the scientific loop.

If you get too skilled at chat-based AI, you become a **copy-paste-run-debug monkey**.



Agentic systems are what happens if you let ChatGPT **edit files, run commands, and inspect the behaviour**.



Case Study 1: Agents do coding grunt work at scale

`emcee` → `BlackJAX`

Will Handley, October 2025
Manual: 2 months. LLM: 15 min.
Matches reference `emcee` numerically.
Written for a Monte Carlo figure in a talk.

`jaxwavelets`

Will Handley
PyWavelets reimplemented in JAX.
10^{-14} agreement, 1,177 tests.
4.7 hours' active computation.
Available on PyPI.

Astro Legacy Archive

Measurements for machine learning.
Proposal under review.
huggingface.co/astro-legacy-archive

`jax-bandflux`

Sam Leeney
[2504.08081]

SALT3 bandflux on GPU: $\sim 100\times$.
Differentiable; initial draft in ~ 3 h.

`JAX-SNANA`

Matt Grayling
SN light-curve pipeline rewritten in JAX

~ 200 simulations/s on CPU
$> 10{,}000$/s on A100.
DES Y5 agreement: < 1 millimag.

Bug Hunting

Sinah Legner: CAMB torsion
N_{eff} counted twice; an evolution pathway for massive neutrinos was missing.

Covariance in minutes

Natalie Hogg: 24 h → 1–2 min.

HPC jobs and meeting scripts

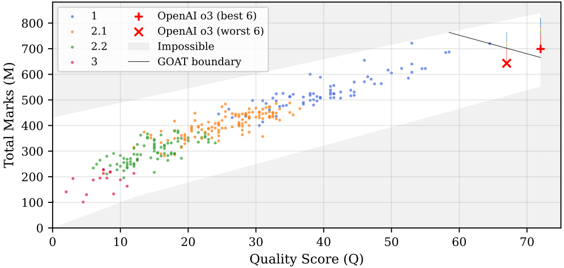
Submit and monitor HPC jobs.
Write and run scripts in meetings.

May 2025

I got OpenAI's o3 to take all four Part II Astrophysics papers.

Toby Lovick transcribed the answers by hand into examination booklets.

I slipped them amongst real scripts for blind marking.



April 2026

I assembled confidential papers locally: Alan with Qwen and Gemma, two RTX 6000s, no internet. Found typos and substantive errors we had missed.

Case Study 3: Live transcripts at SBI4GALEV

Simulation Based Inference for Galaxy Evolution, 23–26 June 2026

Toby Lovick
Built the system

Third meeting in the series, 62 participants.

The local model's account of the science:
SBI moving from tutorials to production. Diffusion and score models replacing flows; differentiable JAX simulators; robustness to misspecification.

Written from its own transcripts.
Byline: "alan (automated rapporteur)".

Chris Lovell
Operated and developed it during the meeting

One participant declined recording but wanted the transcript. We could not separate the two.

Publication required an explicit opt-in.
53 talks captured, 39 in the public archive.

Keeping the data in the room is what makes this acceptable, in my view.
NeMo and Gemma 4 31B on `turing`, our GPU machine. No cloud processing.

Public archive [github:sbi-galev/ambient-ai]

An agent cannot do all the science in a paper

My rule of thumb: about 80% of the work can be handed over

80% Grunt work for agents	20% Human judgement

We decide what to calculate, and what to make of the answer: judgement and taste.

Those choices were human in every example you've just seen.

A paper produced by agents from start to finish leaves that part undone.

In principle: 5× faster, 5× more fun.

AI in our research lab today

Claude Code, Codex, Otter and our GitHub repositories

Otter and `mdrecord`

Otter $\xrightarrow{\text{nightly}}$ `mdrecord`

- About 1,000 meetings saved in 18 months
- This talk began with a search through our meeting transcripts

Claude Code and Codex

Code, slides and tasks during meetings

- Nikku Madhusudhan: K2-18b retrieval code assembled as we talked
- James Alvey and Thomas Spieksma: requested Einstein Telescope figure and source quotations displayed within 15 s

Our 2025 tools: [github:handley-lab/mcp-handley-lab]

The Ambient AI Lab we're building

Cosmos grant: $5,000 for 90 days of development

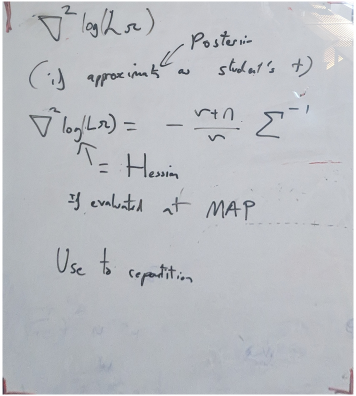
- Frontier tools expect us at a computer, and our work follows.
- We work best together at a board, in meetings and at conferences
- Bring agents into the room before we become digitally rendered meat-backed slack bots
- Record speech and the board without changing how people work
- Rolling out to supervisions in Michaelmas and Lent, as well as to our science

Cosmos Institute:
"Instrumenting the Unrecorded Half of Science"

Checking the whiteboard

Toby Lovick

Photograph → Gemini 3.1 → LaTeX; then the live loop



$$\nabla^2 \log(L\Omega)$$

Posterior

(if approximate as student's t)

$$\underset{\substack{\uparrow \\ =\ \text{Hessian}}}{\nabla^2 \log(L\Omega)} = -\frac{v+n}{v}\Sigma^{-1}$$

If evaluated at MAP

Use to repartition

- Live: Pi camera → turing → projector
- Ask "where is this wrong?"

We do not need frontier AI labs for this work

Local models caught errors the examiners missed

- Qwen and Gemma found typos and substantive mistakes in the 2026 papers that we had missed
- I kept the papers off Claude Code and Codex: I could not risk them appearing in students' prompts
- Local recording made consent possible at SBI4GALEV. Toby's whiteboard runs Gemma.

40× usage

Claude Code and Codex

Frontier vendors sell a harness and subsidised tokens. We can write the harness and run the models ourselves.

The reckoning is coming where the price tag becomes honest.

It's made by scientists for scientists, not by some conference company or some frontier AI tech firm.

Conclusions

- **Agents do grunt work at scale.**
  In principle: $5\times$ more science and $5\times$ more fun.
- **The other 20% is human judgement and taste.**
  Generating a paper does not do that work.
- **This work needs no frontier AI lab.**
  Made by scientists for scientists.
- **The agents should come to the room.**
  People work together at boards, in meetings and at conferences.

Ambient means AI that feels like electricity, not a digital scientist.

Where in your working week would an agent listening to the room be useful?